# Unit-2

# Data Pre-processing

# Data Preprocessing

- Data preprocessing is the process of **transforming raw data into an understandable** format.
- The **quality of the data** should be checked before applying machine learning or data mining algorithms.

## What is the need of Pre-processing ?

- Goal is to improve the quality of the data.
- Helps in handling missing values, removing duplicates, and normalizing data.
- Ensures the accuracy and consistency of the dataset.

Data Integration
Data Cleaning
Data Preprocessing
Data Transformation
Data Reduction or Dimension Reduction
V7 Labs

CO2 emissions (metric tons per capita)

| Country Name | 2000 | 2001 | 2002 |
|---|---|---|---|
| United States | 20.17875 | 19.63651 | NAN |
| United Kingdom | 9.199549 | 9.233175 | 8.904123 |
| India | 0.97987 | NAN | 0.967381 |
| China | 2.696862 | 2.742121 | 3.007083 |
| Russian Federation | NAN | 10.6696 | 10.7159 |
| Australia | 17.20061 | 16.73337 | 17.37045 |

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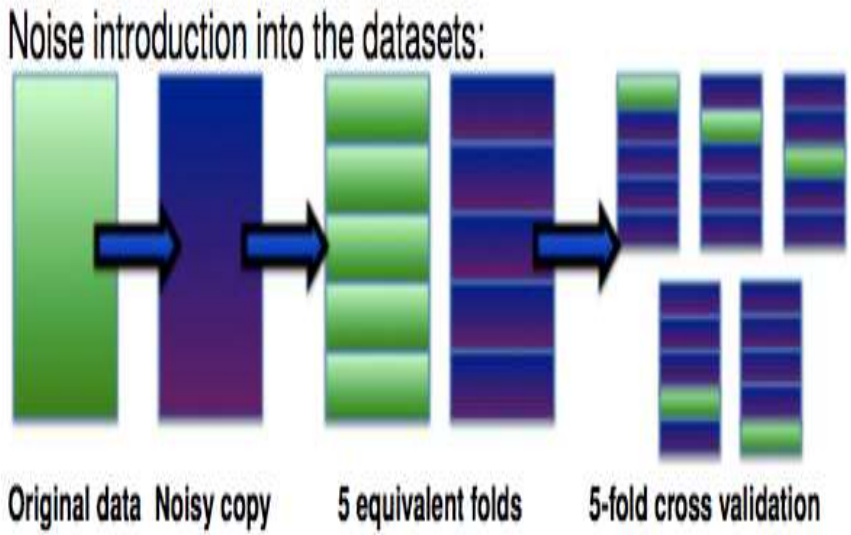
# Data cleaning

1. **Data Cleaning:** It is the process of identifying and correcting errors in the dataset. It involves handling missing values, removing duplicates, and correcting incorrect or outlier data to ensure the dataset is accurate and reliable. Clean data is essential for effective analysis, as it improves the quality of results and enhances the performance of data models.

- **Missing Values:** This occur when data is absent from a dataset. You can either ignore the rows with missing data or fill the gaps manually, with the attribute mean, or by using the most probable value. This ensures the dataset remains accurate and complete for analysis.

- **Noisy Data:**




**How to remove Noise ?**

It refers to irrelevant or incorrect data that is difficult for machines to interpret, often caused by errors in data collection or entry. It can be handled in several ways:  Binning,

# Ways to remove Noise from Data:

- **Binning Method:** The data is sorted into equal segments, and each segment is smoothed by replacing values with the mean or boundary values.

- **Clustering:** This method groups similar data points together, with outliers either being undetected or falling outside the clusters. These techniques help remove noise and improve data quality.

# Data integration

- The process of combining multiple sources into a single dataset. The Data integration process is one of the main components in data management.

# Data Integration

- **Schema integration**: Integrates metadata(a set of data that describes other data) from different sources.
- **Entity identification problem:** Identifying entities from multiple databases.
- **Detecting and resolving data value concepts**: The data taken from different databases while merging  may differ from another database. For example, the date format may differ like “MM/DD/YYYY” or “DD/MM/YYYY”.

# Data Transformation

- The change made in the format or the structure of the data is called data transformation. This step can be simple or complex based on the requirements.

# Data Transformation

- **Normalization:**
  It is done in order to scale the data values in a specified range (-1.0 to 1.0 or 0.0 to 1.0)
- **Attribute Selection:**
  In this strategy, new attributes are constructed from the given set of attributes to help the mining process.

# Data Transformation

- **Discretization**: The continuous data here is split into intervals. interval like (3 pm-5 pm, 6 pm-8 pm).
- **Concept Hierarchy Generation:**
  Here attributes are converted from lower level to higher level in hierarchy. For Example-The attribute “city” can be converted to “country”.

# Data Reduction

- This process helps in the reduction of the volume of the data which makes the analysis easier yet produces the same or almost the same result. This reduction also helps to reduce storage space.

# Data Reduction

- **Attribute Subset Selection:**
  The highly relevant attributes should be used, rest all can be discarded.
- **Numerosity Reduction:**
  This enable to store the model of data instead of whole data, for example: Regression Models.
- **Dimensionality Reduction:**
  This reduce the size of data by encoding mechanisms.It can be lossy or lossless.

# Data Preprocessing: Best practices

- The first step in Data Preprocessing is to understand your data. Just looking at your dataset can give you an intuition of what things you need to focus on.
- Use statistical methods or pre-built libraries that help you visualize the dataset and give a clear image of how your data looks in terms of class distribution.
- Summarize your data in terms of the number of duplicates, missing values, and outliers present in the data.

# Data Preprocessing: Best practices

- Drop the fields you think have no use for the modeling or are closely related to other attributes. Dimensionality reduction is one of the very important aspects of Data Preprocessing.
- Do some feature engineering and figure out which attributes contribute most towards model training.

# Splitting dataset into Training and Testing set.

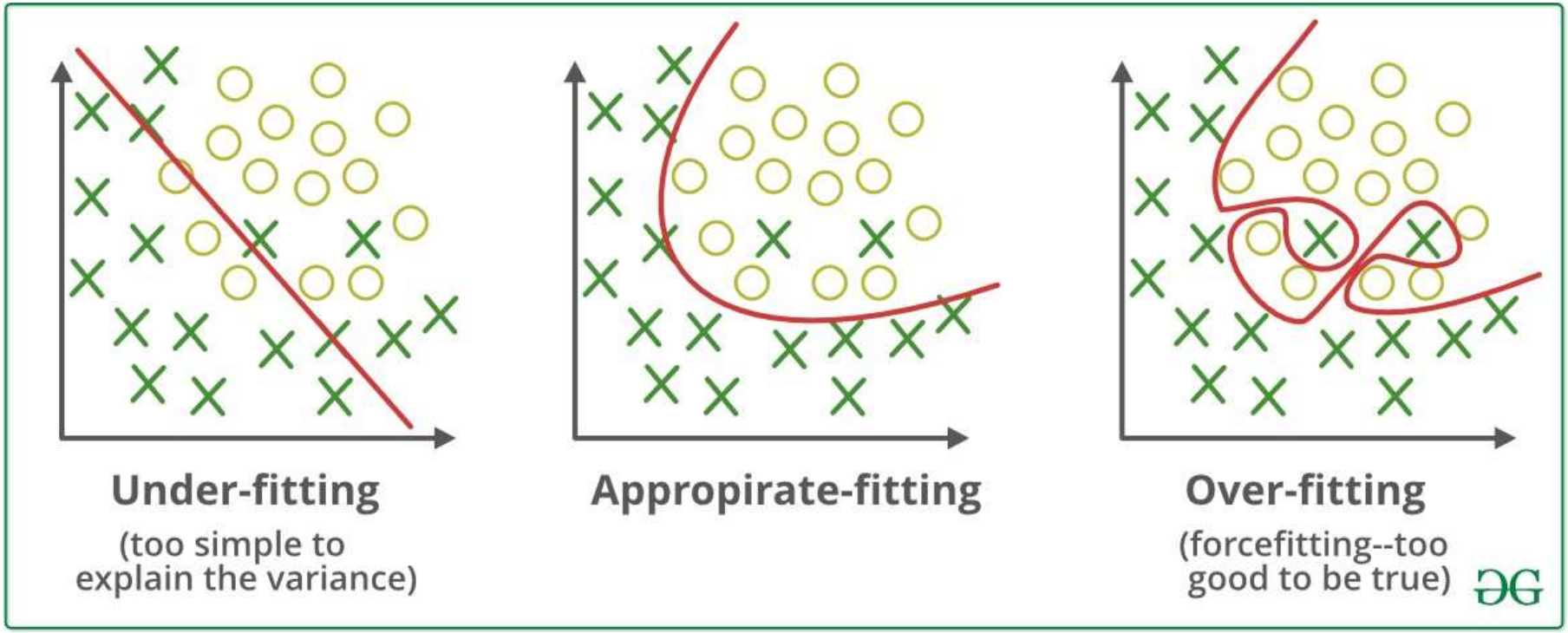
- Training Data
  - The observations in the training set form the experience that the algorithm uses to learn. In supervised learning problems, each observation consists of an observed output variable and one or more observed input variables.
- Test Data
  - The test set is a set of observations used to evaluate the performance of the model using some performance metric. It is important that no observations from the training set are included in the test set.

# Basics

- **Bias:** Assumptions made by a model to make a function easier to learn.
- **Variance:** If you train your data on training data and obtain a very low error, upon changing the data and then training the same previous model you experience a high error, this is variance.

# Under & Over Fitting

- **Underfitting:**
  - A statistical model or a machine learning algorithm is said to have underfitting when it cannot capture the underlying trend of the data.
- **Overfitting:**
  - A statistical model is said to be overfitted when we train it with a lot of data *(just like fitting ourselves in oversized pants!)*.


Under-fitting
(too simple to explain the variance)
Appropirate-fitting
Over-fitting
(forcefitting--too good to be true)

# References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- https://www.v7labs.com/blog/data-preprocessing-guide